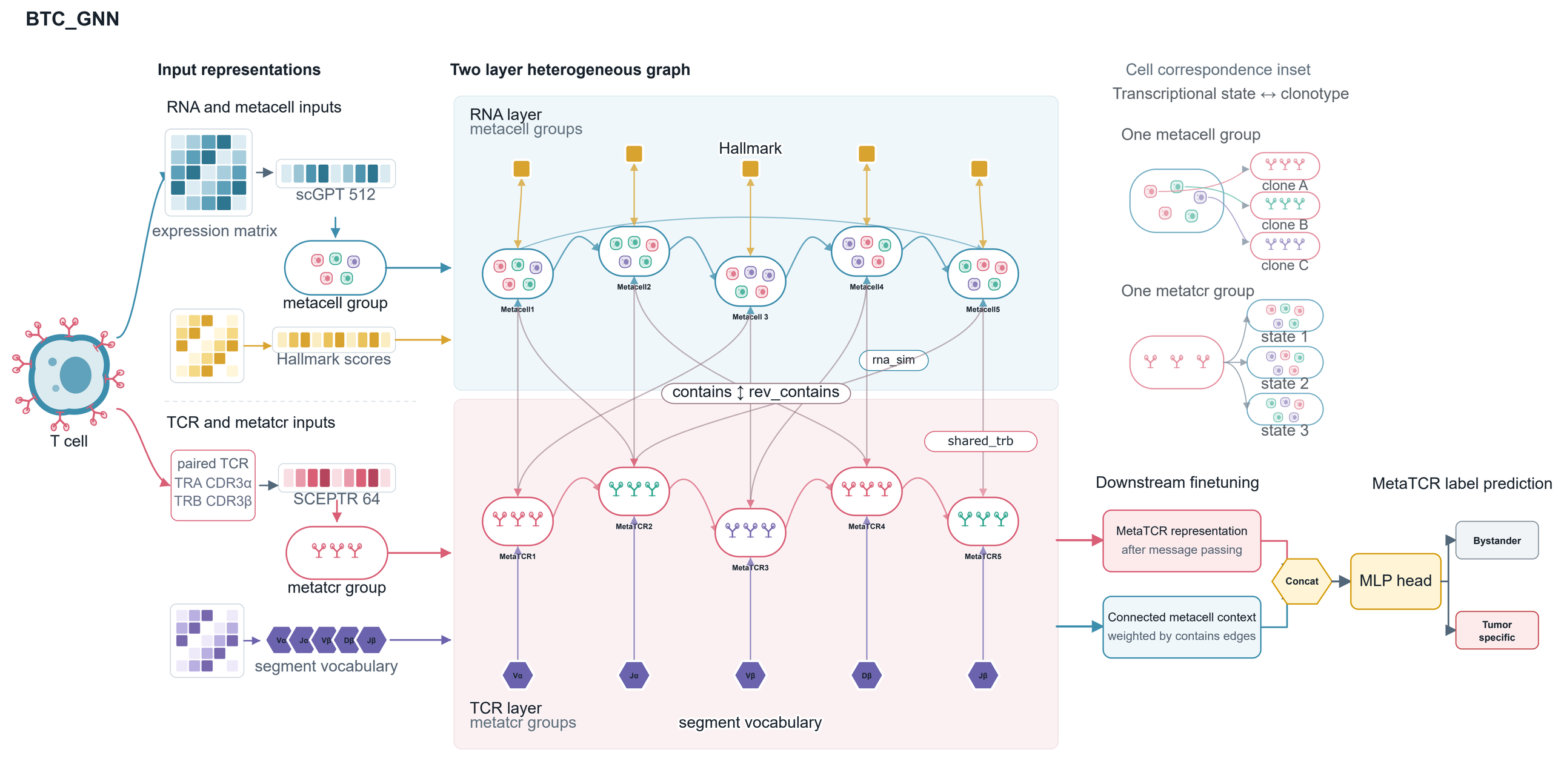

BTC_GNN
Input representations
RNA and metacell inputs
expression matrix
scGPT 512
metacell group
Hallmark scores
T cell
TCR and metatcr inputs
paired TCR
TRA CDR3α
TRB CDR3β
SCEPTR 64
metatcr group
Vα
Jα
Vβ
Dβ
Jβ
segment vocabulary
Two layer heterogeneous graph
RNA layer
metacell groups
Hallmark
Metacell1
Metacell2
Metacell 3
Metacell4
Metacell5
rna_sim
contains ↕ rev_contains
shared_trb
MetaTCR1
MetaTCR2
MetaTCR3
MetaTCR4
MetaTCR5
TCR layer
metatcr groups
segment vocabulary
Cell correspondence inset
Transcriptional state ↔ clonotype
One metacell group
clone A
clone B
clone C
One metatcr group
state 1
state 2
state 3
Downstream finetuning
MetaTCR representation
after message passing
Connected metacell context
weighted by contains edges
Concat
MLP head
MetaTCR label prediction
Bystander
Tumor specific